

COMPUTER SECURITY PRACTICAL REPORT

CHAPTER 3-5(Scanning, Enumeration, Vulnerability Analysis)

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LAB 1.1 – HOST DISCOVERY (PING SWEEP)

Step 1 – Identify your network range.

-In the below picture, a command called ip a was used to display the ip range of the network that is in the target machine.

```
(kali㉿kali)-[~]  
$ ip a  
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
inet 127.0.0.1/8 scope host lo  
    valid_lft forever preferred_lft forever  
inet6 ::1/128 scope host noprefixroute  
    valid_lft forever preferred_lft forever  
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
link/ether 08:00:27:81:2f:ed brd ff:ff:ff:ff:ff:ff  
inet 192.168.56.101/24 brd 192.168.56.255 scope global noprefixroute eth0  
    valid_lft forever preferred_lft forever  
inet6 fe80::b06f:494c:c311:a0c5/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever  
: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
link/ether 08:00:27:cd:53:cd brd ff:ff:ff:ff:ff:ff  
inet 10.0.4.15/24 brd 10.0.4.255 scope global dynamic noprefixroute eth1  
    valid_lft 86282sec preferred_lft 86282sec  
inet6 fd17:625c:f037:4:ccba:fe22:3ce9:7412/64 scope global dynamic nopref  
    valid_lft 86284sec preferred_lft 14284sec  
inet6 fe80::6df2:99e8:1581:96a5/64 scope link noprefixroute  
    valid_lft forever preferred_lft forever
```

Step 2 – Perform a Ping Sweep

-In this step fping -a -g command was entered which also included the network range using the ping sweep to discover the number of host which were alive in the network. It returned the ip addresses of the hosts that were alive in this network.

```

(kali@kali)-[~]
$ fping -a -g 192.168.56.0/24
192.168.56.1
192.168.56.101
192.168.56.102
192.168.56.103
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.4
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.4
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.4
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.4
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.3
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.3
ICMP Host Unreachable from 192.168.56.101 for ICMP Echo sent to 192.168.56.3

```

Step 3 – Verify Results

-In this step, it was the confirmation of the previous step configured using ping sweep to confirm if the host of the specified ip address is alive.

```

(kali@kali)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.477 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=0.529 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.34 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=0.632 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=1.21 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=0.723 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=0.739 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=0.647 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=0.765 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=0.767 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=0.731 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=0.733 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=0.730 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=0.727 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=0.746 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=0.734 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=0.590 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=0.747 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=0.758 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=0.743 ms
64 bytes from 192.168.56.102: icmp_seq=21 ttl=64 time=0.776 ms
64 bytes from 192.168.56.102: icmp_seq=22 ttl=64 time=0.739 ms
64 bytes from 192.168.56.102: icmp_seq=23 ttl=64 time=0.740 ms
64 bytes from 192.168.56.102: icmp_seq=24 ttl=64 time=0.740 ms
64 bytes from 192.168.56.102: icmp_seq=25 ttl=64 time=0.726 ms
64 bytes from 192.168.56.102: icmp_seq=26 ttl=64 time=0.757 ms
64 bytes from 192.168.56.102: icmp_seq=27 ttl=64 time=0.720 ms
64 bytes from 192.168.56.102: icmp_seq=28 ttl=64 time=0.746 ms

```

Step 4 – Document Findings

Host	Status
192.168.56.101	Kali
192.168.56.102	Metasploitable
192.156.56.103	Windows

Student Reflection

Why ICMP is used in host discovery.

-It is because hosts mostly use it to reply or confirm that they can be reached by sending echo replies . This makes it easier for attackers to see if devices are alive in a network.

Why some hosts may block ICMP.

-It because it reveals information about if the host is alive or not alive in a network.

-To prevent attacks from taking place.

LAB 1.2 – PORT SCANNING

Step 1 – Basic Port Scan

-In this step Nmap was used to discover whether ports are open or closed.

```
(kali㉿kali)-[~]  
$ nmap 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-10 14:31 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00058s latency).  
Not shown: 980 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
MAC Address: 08:00:27:6C:71:B4 (Oracle VirtualBox virtual NIC)
```

Step 2 – Full Port Scan

-In this step nmap -p- command was used on Nmap to scan all the available ports in this network.

```
(kali@kali)-[~]  
$ nmap -p- 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-10 14:33 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.00053s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
3632/tcp  open  distccd  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
6697/tcp  open  ircs-u  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
8787/tcp  open  msgsrvr  
55832/tcp open  unknown
```



```
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
55832/tcp open  unknown
57074/tcp open  unknown
58057/tcp open  unknown
60054/tcp open  unknown
MAC Address: 08:00:27:6C:71:B4 (Oracle VirtualBox virtual NIC)
```

Step 3 – TCP SYN Scan

-In this step, a stealth scanning was ran to find a open port without completing the full TCP three way handshake .

```
L$ nmap -sS 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-10 14:36 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00082s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:6C:71:B4 (Oracle VirtualBox virtual NIC)
```

Step 4 – TCP Connect Scan

-In this step , a port scan was ran and unlike the previous step this one completed the three way handshake .


```
L$ nmap -sT 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-10 14:39 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:6C:71:B4 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.49 seconds
```

Step 5 – UDP Scan

- In this step Nmap was used to conduct a udp port scan and it was discovered that the udp scan was very slow because it did not confirm connections.

```
(kali@kali)-[~]  
$ nmap -sU 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-10 14:40 -0400  
Stats: 0:06:36 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 44.65% done; ETC: 14:55 (0:08:08 remaining)  
Stats: 0:06:42 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 45.27% done; ETC: 14:55 (0:08:04 remaining)  
Stats: 0:07:02 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 47.25% done; ETC: 14:55 (0:07:49 remaining)  
Stats: 0:07:05 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 47.37% done; ETC: 14:55 (0:07:50 remaining)  
Stats: 0:09:32 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 62.08% done; ETC: 14:56 (0:05:48 remaining)  
Stats: 0:13:04 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 82.53% done; ETC: 14:56 (0:02:45 remaining)  
Stats: 0:15:09 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 94.53% done; ETC: 14:56 (0:00:52 remaining)  
Stats: 0:15:53 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 98.82% done; ETC: 14:56 (0:00:11 remaining)  
Stats: 0:16:15 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 99.99% done; ETC: 14:57 (0:00:00 remaining)  
Stats: 0:16:21 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 99.99% done; ETC: 14:57 (0:00:00 remaining)  
Stats: 0:16:25 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan  
UDP Scan Timing: About 99.99% done; ETC: 14:57 (0:00:00 remaining)
```

LAB 1.3 – SERVICE VERSION DETECTION

-In this step Nmap was used to identify the software running on a port which helps to identify known vulnerabilities.

Session Actions Edit View Help

(kali@kali)-[~]

\$ nmap -sV 192.168.56.102

Starting Nmap 7.98 (<https://nmap.org>) at 2026-03-11 08:32 -0400

Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan

Service scan Timing: About 91.30% done; ETC: 08:33 (0:00:02 remaining)

Stats: 0:00:42 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan

Service scan Timing: About 95.65% done; ETC: 08:33 (0:00:02 remaining)

Nmap scan report for 192.168.56.102

Host is up (0.00069s latency).

Not shown: 977 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login?	
514/tcp	open	shell	Netkit rshd
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	(access denied)

PORT	STATE	SERVICE	VERSION
21/tcp	open	ftp	vsftpd 2.3.4
22/tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
25/tcp	open	smtp	Postfix smtpd
53/tcp	open	domain	ISC BIND 9.4.2
80/tcp	open	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	2 (RPC #100000)
139/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec	netkit-rsh rexecd
513/tcp	open	login?	
514/tcp	open	shell	Netkit rshd
1099/tcp	open	java-rmi	GNU Classpath grmiregistry
1524/tcp	open	bindshell	Metasploitable root shell
2049/tcp	open	nfs	2-4 (RPC #100003)
2121/tcp	open	ftp	ProFTPD 1.3.1
3306/tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	VNC (protocol 3.3)
6000/tcp	open	X11	(access denied)

6667/tcp open irc UnrealIRCd

8009/tcp open ajp13 Apache Jserv (Protocol v1.3)

8180/tcp open http Apache Tomcat/Coyote JSP engine 1.1

MAC Address: 08:00:27:6C:71:B4 (Oracle VirtualBox virtual NIC)

Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 58.35 seconds

LAB 1.4 – OPERATING SYSTEM DETECTION

-In this step nmap was used to identify the os family of the target machine .

```
(kali㉿kali)-[~]  
$ nmap -O 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 08:43 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0015s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown
```

LAB 1.5 – NMAP SCRIPTING ENGINE

-In the step an automated script on Nmap was used to collect advanced information like the SMB information, SSL certificates and DNS data.


```

(kali@kali) [~]
$ nmap --script=discovery 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 08:53 -0400
No profinet devices in the subnet
Pre-scan script results:
| targets-ipv6-multicast-echo:
|   IP: fe80::a00:27ff:fe6c:71b4  MAC: 08:00:27:6c:71:b4  IFACE: eth0
|   Use --script-args=newtargets to add the results as targets
| targets-ipv6-multicast-mld:
|   IP: fe80::a00:27ff:fe6c:71b4  MAC: 08:00:27:6c:71:b4  IFACE: eth0
|   IP: fe80::ccf4:d4d6:9a95:9102  MAC: 08:00:27:c4:24:07  IFACE: eth0
|   IP: fe80::ed96:6b4e:a318:a6be  MAC: 0a:00:27:00:00:11  IFACE: eth0
|
|   Use --script-args=newtargets to add the results as targets
| targets-sniffer: Sniffed 18 address(es).
| fe80::b06f:494c:c311:a0c5
| fe80::ed96:6b4e:a318:a6be
| fe80::a00:27ff:fe6c:71b4
| fe80::ccf4:d4d6:9a95:9102
| fe80::a00:27ff:fe81:2fed
| 192.168.56.0
| 224.0.0.2
| 192.168.56.102
| 192.168.56.103
| ::
| 224.0.0.252
| c0a8:3865::
| 224.0.0.251
| 239.255.255.250

```

LAB 2 – ENUMERATION (CHAPTER 4)

LAB 2.1 – SMB ENUMERATION

-In this step , enum4linux-ng tool was used to collect information about the windows and the information that was extracted is users, shares etc.

```

(kali㉿kali)-[~]
(kali㉿kali)-[~]
$ enum4linux -ng 192.168.56.102
Unknown option: g
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on We
d Mar 11 08:58:16 2026

===== ( Target Information ) =====
=====
Target ..... 192.168.56.102
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.56.102 ) =====
=====

[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.56.102 ) =====
=====

Looking up status of 192.168.56.102

```

```

Home Trash
===== ( Enumerating Workgroup/Domain on 192.168.56.102 ) =====
=====

[+] Got domain/workgroup name: WORKGROUP

File System
===== ( Nbtstat Information for 192.168.56.102 ) =====
=====

Looking up status of 192.168.56.102
METASPLOITABLE <00> - B <ACTIVE> Workstation Service
METASPLOITABLE <03> - B <ACTIVE> Messenger Service
METASPLOITABLE <20> - B <ACTIVE> File Server Service
.._MSBROWSE_.. <01> - <GROUP> B <ACTIVE> Master Browser
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
WORKGROUP <1d> - B <ACTIVE> Master Browser
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections

MAC Address = 00-00-00-00-00-00

===== ( Session Check on 192.168.56.102 ) =====
=====

[+] Server 192.168.56.102 allows sessions using username '', password ''

===== ( Getting domain SID for 192.168.56.102 ) =====

```

LAB 2.2 – SHARE ENUMERATION

-In this step a smbclient -L //192.168.56.103 command was used but access was denied so to gain access to the folders shared ,

the -U 222932050(windows user) was implemented into the command.

```
(kali㉿kali)-[~]
$ smbclient -L //192.168.56.103

Password for [WORKGROUP\kali]:
session setup failed: NT_STATUS_ACCESS_DENIED

(kali㉿kali)-[~]
$ smbclient -L //192.168.56.103 -U 222932050

Password for [WORKGROUP\222932050]:

      Sharename      Type      Comment
      ────
      ADMIN$         Disk      Remote Admin
      C$              Disk      Default share
      IPC$            IPC       Remote IPC
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.56.103 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali㉿kali)-[~]
$
```

LAB 2.3 – SNMP ENUMERATION

-The `snmpwalk -v2c -c public 192.168.56.102` command was used to gather information about the network devices and servers but unfortunately the information could not be obtained because the port 161 could not be accessed on the target machine.

```
(kali㉿kali)-[~]
$ snmpwalk -v2c -c public 192.168.56.102
Timeout: No Response from 192.168.56.102

(kali㉿kali)-[~]
$ nmap -sU -p 161 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-12 05:04 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00055s latency).

PORT      STATE SERVICE
161/udp   closed snmp
MAC Address: 08:00:27:F4:37:E1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.08 seconds
```

LAB 2.4 – DNS ENUMERATION

-In this step the main goal was to retrieve mail servers .

```
(kali㉿kali)-[~]
$ dig example.com MX

; <<>> DiG 9.20.18-1-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 48127
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 4000
;; QUESTION SECTION:
;example.com.                IN      MX

;; ANSWER SECTION:
example.com.                267     IN      MX      0 .

;; Query time: 27 msec
;; SERVER: 168.172.121.1#53(168.172.121.1) (UDP)
;; WHEN: Thu Mar 12 04:51:52 EDT 2026
;; MSG SIZE rcvd: 55

(kali㉿kali)-[~]
$
```

LAB 2.5 – SMTP ENUMERATION

-In this step the command nc -nv was entered to determine the server software and the hostname.

```
(kali㉿kali)-[~]
$ nc -nv 192.168.56.102 25
(UNKNOWN) [192.168.56.102] 25 (smtp) open
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
```

LAB 2.6 – WEB ENUMERATION

-In these steps commands were used to check for headers and others were used to run finger printing .

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ curl -I http://192.168.56.102  
HTTP/1.1 200 OK  
Date: Thu, 12 Mar 2026 00:43:50 GMT  
Server: Apache/2.2.8 (Ubuntu) DAV/2  
X-Powered-By: PHP/5.2.4-2ubuntu5.10  
Content-Type: text/html  
(kali@kali)-[~]
```

```
(kali@kali)-[~]  
$ whatweb http://192.168.56.102  
http://192.168.56.102 [200 OK] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu  
][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.56.102], PHP[5.2.4-2ubuntu5.10], Title[Meta  
able2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]  
(kali@kali)-[~]  
$
```

LAB 3.1 – NMAP VULNERABILITY SCAN

```
$ nmap --script=discovery 192.168.56.102  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-12 06:57 -0400  
No profinet devices in the subnet  
Pre-scan script results:  
| broadcast-igmp-discovery:  
| 192.168.56.0  
|   Interface: eth0  
|   Version: 2  
|   Group: 224.0.0.252  
|   Description: Link-local Multicast Name Resolution (rfc4795)  
| 192.168.56.103  
|   Interface: eth0  
|   Version: 2  
|   Group: 224.0.0.251  
|   Description: mDNS (rfc6762)  
| 192.168.56.103  
|   Interface: eth0  
|   Version: 2  
|   Group: 224.0.0.252  
|   Description: Link-local Multicast Name Resolution (rfc4795)  
| 192.168.56.0  
|   Interface: eth0  
|   Version: 2  
|   Group: 239.255.255.250  
|   Description: Organization-Local Scope (rfc2365)  
| Use the newtargets script-arg to add the results as targets  
|_ ipv6-multicast-mld-list:  
|   fe80::a00:27ff:fe6c:71b4:  
|     device: eth0  
|     mac: 08:00:27:6c:71:b4  
|     multicast_ips:  
|       ff02::1:ff6c:71b4 (NDP Solicited-node)
```

LAB 3.2 – NIKTO WEB VULNERABILITY SCAN

- In this step nikto vulnerability scan was used to find outdated systems from the target website .

```
(kali㉿kali)-[~]
$ nikto -h http://192.168.56.102
- Nikto v2.5.0

+ Target IP: 192.168.56.102
+ Target Hostname: 192.168.56.102
+ Target Port: 80
+ Start Time: 2026-03-12 07:03:00 (GMT-4)

+ Server: Apache/2.2.8 (Ubuntu) DAV/2
+ /: Retrieved x-powered-by header: PHP/5.2.4-2ubuntu5.10.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://develop
illa.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to
the content of the site in a different fashion to the MIME type. See: https://www.net
r.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ /index: Uncommon header 'tcn' found, with contents: list.
+ /index: Apache mod_negotiation is enabled with MultiViews, which allows attackers to
y brute force file names. The following alternatives for 'index' were found: index.php
http://www.wisec.it/sectou.php?id=4698ebdc59d15,https://exchange.xforce.ibmcloud.com/
abilities/8275
+ Apache/2.2.8 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.
the EOL for the 2.x branch.
+ /: Web Server returns a valid response with junk HTTP methods which may cause false
ves.
+ /: HTTP TRACE method is active which suggests the host is vulnerable to XST. See: ht
owasp.org/www-community/attacks/Cross_Site_Tracing
+ /phpinfo.php: Output from the phpinfo() function was found.
+ /doc/: Directory indexing found.
+ /doc/: The /doc/ directory is browsable. This may be /usr/doc. See: http://cve.mitre
gi-bin/cvename.cgi?name=CVE-1999-0678
+ /?=PHPB8B5F2A0-3C92-11d3-A3A9-4C7B08C10000: PHP reveals potentially sensitive inform
```

Extra Lab :Vulnerability Assessment Using Nessus

6. Step 1 -Install Nessus

-In this step Nessus Essentials tool was installed which is a tool that is used to detect security weaknesses in servers .

```

(kali㉿kali)-[~/Downloads]
$ sudo dpkg -i Nessus-*.deb 0.102) 56(84) bytes of data.
Selecting previously unselected package nessus. time=1.57 ms
(Reading database... 444549 files and directories currently installed.)
Preparing to unpack Nessus-10.11.3-ubuntu1604_amd64.deb...09 ms
Unpacking nessus (10.11.3)...
Setting up nessus (10.11.3)... files ---
HMAC : (Module_Integrity) : Pass | 0% packet loss, time 2028ms
SHA1 : (KAT_Digest) : Pass | 2/0.785/1.566/0.551 ms
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KBKDF : (KAT_KDF) : Pass
HKDF : (KAT_KDF) : Pass
SSKDF : (KAT_KDF) : Pass
X963KDF : (KAT_KDF) : Pass
X942KDF : (KAT_KDF) : Pass
HASH : (DRBG) : Pass
CTR : (DRBG) : Pass
HMAC : (DRBG) : Pass

```

-In this step a command was entered to check if the Nessus is active and it was a success.

```

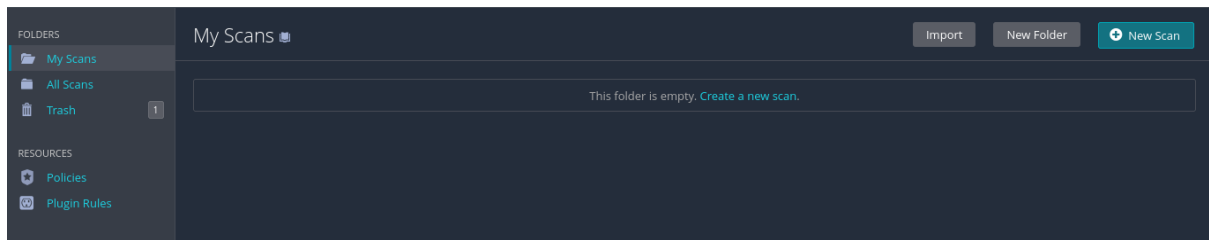
(kali㉿kali)-[~/Downloads]
$ sudo systemctl status nessusd
[sudo] password for kali:
● nessusd.service - The Nessus Vulnerability Scanner
   Loaded: loaded (/usr/lib/systemd/system/nessusd.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-03-13 12:11:31 EDT; 43min ago
  Invocation: 284752409ca44fc58a0f3638939f45ac
    Main PID: 44629 (nessus-service)
      Tasks: 17 (limit: 4456)
     Memory: 1.3G (peak: 1.6G, swap: 116K, swap peak: 9.8M)
        CPU: 31min 33.894s
    CGroup: /system.slice/nessusd.service
            └─44629 /opt/nessus/sbin/nessus-service -q
              └─50341 nessusd -q

Mar 13 12:11:31 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.
Mar 13 12:11:31 kali nessus-service[44629]: nessus-service [44629][INFO] : Nessus 19.16.3 [build 20047] Started

```

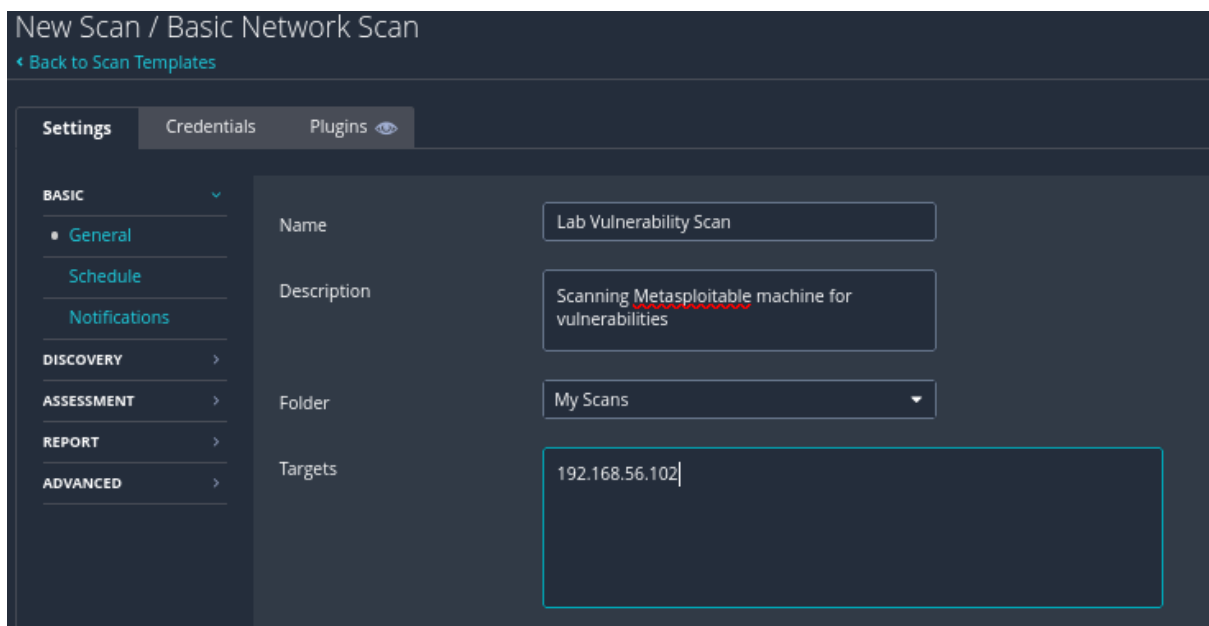
7. Step 2 - Access Nessus Web Interface

-In this step a Nessus account was created by the user .



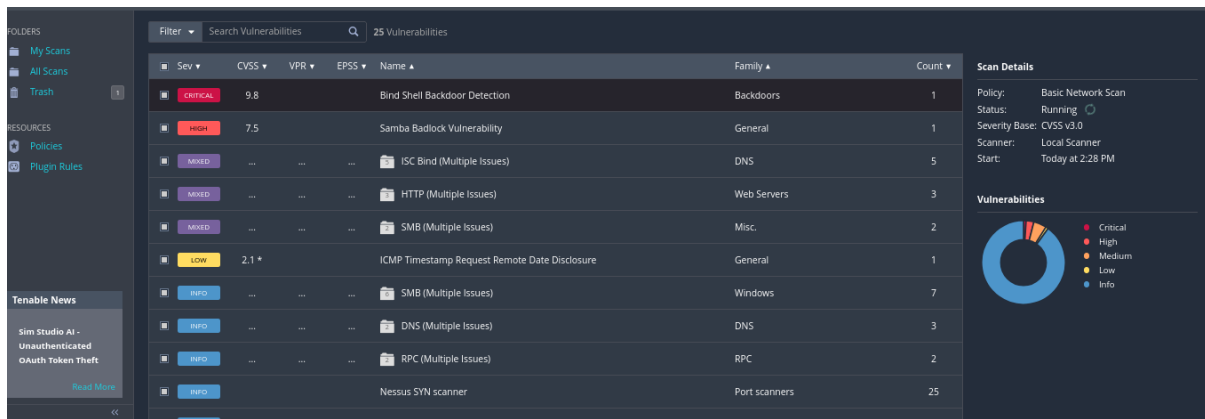
8. Step 3 - Create a New Scan & 9. Step 4 -Configure the Scan

-In these steps it was a new scan on Nessus had to be created. The Nessus scan was target at one of the machine labs that were used for testing which is the Metasploit able machine .



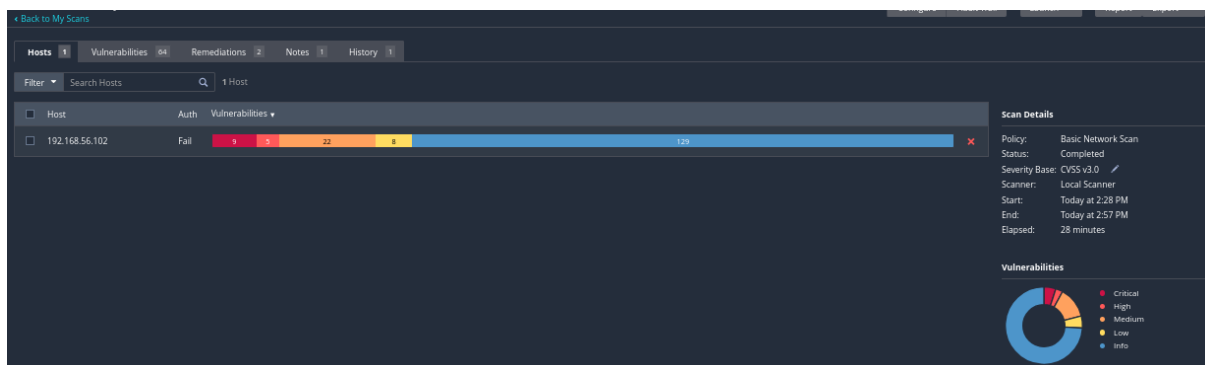
10. Step 5 - Launch the Scan& 11.Step 6 -Monitor the Scan

-The network scan against metasploitable was launched , it was done to monitor things like port scanning and service detection.



12. Step 7 – View Scan Results

-After 28 minutes the scan was completed and it was found out that there was a low chance of exploitation and a higher chance of no risk.



13. Step 8 - Analyse a Vulnerability

- the Canonical ubuntu Linux was detected to chosen to be viewed as the vulnerability and it was found that the version is not maintained by its vendor and that may cause for the security patches to not be released and that may cause security vulnerabilities. The solution to this for the machine to upgrade to a its canonical ubuntu Linux.

CRITICAL

Canonical Ubuntu Linux SEoL (8.04.x)

Description

According to its version, Canonical Ubuntu Linux is 8.04.x. It is, therefore, no longer maintained by its vendor or provider.

Lack of support implies that no new security patches for the product will be released by the vendor. As a result, it may contain security vulnerabilities.

Solution

Upgrade to a version of Canonical Ubuntu Linux that is currently supported.

See Also

<http://www.nessus.org/u73bdb2d2e>

Output

```
OS : Ubuntu Linux 8.04
Security End of Life : May 9, 2013
Time since Security End of Life (Est.) : >= 12 years
```

To see debug logs, please visit individual host

Plugin Details

Severity: Critical
ID: 201352
Version: 1.2
Type: combined
Family: General
Published: July 3, 2024
Modified: March 26, 2025

Risk Information

Risk Factor: Critical
CVSS v3.0 Base Score: 10.0
CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:C/C:H/I:W/A:H
CVSS v2.0 Base Score: 10.0
CVSS v2.0 Vector: CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

14. Step 9 -Review CVE Information

-In this step the CVE was that was identified in the target machine was the type of server used by the machine .

Hosts 1

Vulnerabilities 64

Remediations 2

Notes 1

History 1

LOW

X Server Detection

Description

The remote host is running an X11 server. X11 is a client-server protocol that can be used to display graphical applications running on a given host on a remote client.

Since the X11 traffic is not ciphered, it is possible for an attacker to eavesdrop on the connection.

Solution

Restrict access to this port. If the X11 client/server facility is not used, disable TCP support in X11 entirely (-nolisten tcp).

Output

```
X11 Version : 11.0
```

To see debug logs, please visit individual host

Port 4

Hosts

Plugin Details

Severity: Low
ID: 10407
Version: 1.38
Type: remote
Family: Service detection
Published: May 12, 2000
Modified: March 5, 2019

Risk Information

Risk Factor: Low
CVSS v2.0 Base Score: 2.4
CVSS v2.0 Vector: CVSS2#AV:N/AC:H/Au:N/C:P/I:N/A:N

15. Step 10 -Generate a Report

-Unfortunately the report could not be exported because exports are not for free users .

Hosts 1

Vulnerabilities 64

Remediations 2

Notes 1

History 1

Filter

Search Vulnerabilities

64 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Count
CRITICAL	10.0			Canonical Ubuntu Linux	1
CRITICAL	10.0			VNC Server 'password'	1
CRITICAL	9.8	8.9	0.9447	Apache Tomcat AJP	1
CRITICAL	9.8			SSL Version 2 and 3 Protocol Detection	2
CRITICAL	9.8			Bind Shell Backdoor Detection	1
HIGH	7.5	5.9	0.7714	Samba Badlock Vulnerability	1
HIGH	7.5			NFS Shares World Readable	1
MEDIUM	6.5			TLS Version 1.0 Protocol Detection	2

Exports are not available in Essentials

Exports are not available in Essentials. Upgrade to Nessus Professional to enable this feature.

Ok

Scan Details

Policy: Basic Network Scan
Status: Completed
Severity Base: CVSS v3.0
Scanner: Local Scanner
Start: Today at 2:28 PM
End: Today at 2:57 PM
Elapsed: 28 minutes

Vulnerabilities

CRITICAL

HIGH

MEDIUM

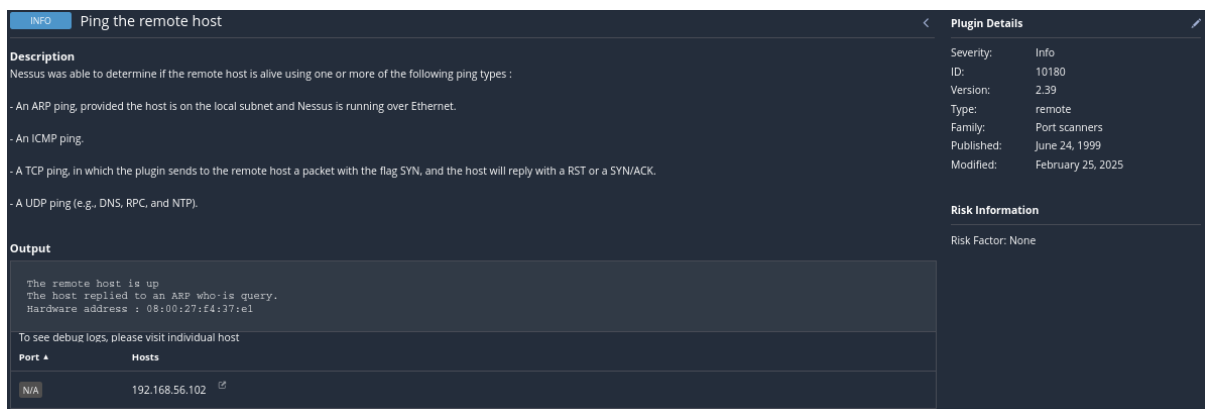
LOW

INFO

16 UNDERSTANDING THE NESSUS SCANNING PROCESS

Stage 1 -Host Discovery

-In this step Nessus was used to determine if the target machine was alive and this was confirmed to be true .Unfortunately the other steps such as port scanning, service detection and vulnerability testing couldn't be performed because of the free version that was being used here .



17. EXAMPLE VULNERABILITIES FOUND

LAB 3.4 – CVE RESEARCH

-In this step a cve research was conducted and the output was that the vulnerability received a base score of 9.8 which shows that it is very critical.

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

vsftpd 2.3.4 downloaded between 20110630 and 20110703 contains a backdoor which opens a shell on port 6200/tcp.

Metrics

CVSS Version 4.0

CVSS Version 3.x

CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:



NIST: NVD

Base Score: 9.8 CRITICAL

Vector: CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H

LAB 3.5 – AI-ASSISTED VULNERABILITY ANALYSIS

LAB 4 : Exploiting SNMP Vulnerabilities and Performing DNS Attacks

PART 1 -SNMP VULNERABILITY EXPLOITATION

7. Step 1 - Identify SNMP Service

-In this step a command was entered to scan the target system using nmap to confirm if the SNMP services is running .The result was the snmp service is not running because the status was closed .

```
(kali㉿kali)-[~] The Nessus Vulnerability Scanner
$ nmap -sU -p 161 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 15:56 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0076s latency).
PORT      STATE SERVICE
161/udp   closed snmp
MAC Address: 08:00:27:F4:37:E1 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 2.61 seconds

(kali㉿kali)-[~] systemd[1]: Started nessusd.service - The Nessus
$
```


8. Step 2 - Enumerate SNMP Using SNMPWalk

-In this step the snmpwalk -v2c – public 192.168.56.102 to find information about the device identity and the system uptime but there was no response from the target machine .

```
(kali㉿kali)-[~]
$ snmpwalk -v2c -c public 192.168.56.102
Timeout: No Response from 192.168.56.102

(kali㉿kali)-[~]
$
```

9. Step 3 - Extract Network Information

```
(kali㉿kali)-[~]
$ snmpwalk -v2c -c public 192.168.56.102 system
system: Unknown Object Identifier (Sub-id not found: (top) → system)
```

10. Step 4 - Use Nmap SNMP Scripts

-In this step a Nmap scripts was ran in order in order to analyse the type of server used by target machine but unfortunately the results did not come as expected because snmp is not running .

```
(kali㉿kali)-[~]
$ nmap -sU -p 161 --script snmp-info 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:07 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00098s latency).

PORT      STATE SERVICE
161/udp    closed snmp
MAC Address: 08:00:27:F4:37:E1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.80 seconds
```

PART 2 - DNS ATTACK LAB

14. Step 1 - Identify DNS Server

-In this step nmap was used to identify the DNS and the output simply indicates the DNS server.

```
(kali@kali)-[~]
$ nmap -p 53 192.168.56.102
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-13 16:09 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00080s latency).

PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 08:00:27:F4:37:E1 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds
```

15. Step 2 — Perform DNS Enumeration

-dns recon command was used to extract dns records and this operation was a success because the output shows A records, MX records and NS records.

```
(kali@kali)-[~]
$ dnsrecon -d example.com
2026-03-13T16:10:19.307634-0400 INFO Starting enumeration for domain: example.com
2026-03-13T16:10:19.308413-0400 INFO std: Performing General Enumeration against: example.com...
2026-03-13T16:10:19.344005-0400 INFO DNSSEC is configured for example.com
2026-03-13T16:10:19.361692-0400 INFO DNSKEYs:
2026-03-13T16:10:19.362301-0400 INFO NSEC ZSK ECDSAP256SHA256 323c9989e94fd06aa7888d7e8fec001b fdedd089830c8963d42c51d28a1b9194 0748a1f616ac7ab66d8d718bb9ecf389 f3
74b30962e4a888d91d2d66879c1b6b
2026-03-13T16:10:19.362582-0400 INFO NSEC KSK ECDSAP256SHA256 99db2cc14cabdc33d6d77da63a2f15f7 1112584f234e8d1dc428e39e8a4a97e1 aa271a555dc90701e17e2a4c4b6f120b 7c
32d44f4ac02bd894cf2d4be7778a19
2026-03-13T16:10:19.362817-0400 INFO NSEC ZSK ECDSAP256SHA256 9318a98e821b3590ec5aa10a698686ab a7ccfd74e1ad1a75b9ee8057d47f9f77 96c691827825896f8ecf104123fd0e95 01
8156ffb505a39997a197f129865adc
2026-03-13T16:10:19.364349-0400 INFO NSEC ZSK ECDSAP256SHA256 a09311112cf9138818cd2feae970ebbd 4d6a30f6088c25b325a39abb5cd1197 aa098283e5aaf421177c2aa5d714992a 99
57d1bcc18f98cd71f1f1806b65e148
2026-03-13T16:10:19.511245-0400 INFO SOA elliottns.cloudflare.com 162.159.44.228
2026-03-13T16:10:19.512042-0400 INFO SOA elliottns.cloudflare.com 108.162.195.228
2026-03-13T16:10:19.512683-0400 INFO SOA elliottns.cloudflare.com 172.64.35.228
2026-03-13T16:10:19.513438-0400 INFO SOA elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
2026-03-13T16:10:19.514314-0400 INFO SOA elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
2026-03-13T16:10:19.514859-0400 INFO SOA elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
2026-03-13T16:10:19.606294-0400 INFO NS hera.ns.cloudflare.com 108.162.192.162
2026-03-13T16:10:19.632883-0400 INFO Bind Version for 108.162.192.162 "2026.2.2"
2026-03-13T16:10:19.634810-0400 INFO NS hera.ns.cloudflare.com 172.64.32.162
2026-03-13T16:10:19.773908-0400 INFO Bind Version for 172.64.32.162 "2026.2.2"
2026-03-13T16:10:19.774274-0400 INFO NS hera.ns.cloudflare.com 173.245.58.162
2026-03-13T16:10:19.883260-0400 INFO Bind Version for 173.245.58.162 "2026.2.2"
2026-03-13T16:10:19.884663-0400 INFO NS hera.ns.cloudflare.com 2803:f800:50::6ca2:c0a2
2026-03-13T16:10:25.899265-0400 INFO NS hera.ns.cloudflare.com 2a06:98c1:50::ac40:20a2
2026-03-13T16:10:31.914944-0400 INFO NS hera.ns.cloudflare.com 2606:4700:50::adf5:3aa2
2026-03-13T16:10:37.945954-0400 INFO NS elliottns.cloudflare.com 172.64.35.228
2026-03-13T16:10:37.978534-0400 INFO Bind Version for 172.64.35.228 "2026.2.2"
```

```
2026-03-13T16:10:37.945954-0400 INFO NS elliottns.cloudflare.com 172.64.35.228
2026-03-13T16:10:37.978534-0400 INFO Bind Version for 172.64.35.228 "2026.2.2"
2026-03-13T16:10:37.979555-0400 INFO NS elliottns.cloudflare.com 162.159.44.228
2026-03-13T16:10:38.116842-0400 INFO Bind Version for 162.159.44.228 "2026.2.2"
2026-03-13T16:10:38.118708-0400 INFO NS elliottns.cloudflare.com 108.162.195.228
2026-03-13T16:10:38.154672-0400 INFO Bind Version for 108.162.195.228 "2026.2.2"
2026-03-13T16:10:38.155152-0400 INFO NS elliottns.cloudflare.com 2803:f800:50::6ca2:c3e4
2026-03-13T16:10:44.174225-0400 INFO NS elliottns.cloudflare.com 2606:4700:58::a29f:2ce4
2026-03-13T16:10:50.182396-0400 INFO NS elliottns.cloudflare.com 2a06:98c1:50::ac40:23e4
2026-03-13T16:10:56.440987-0400 INFO A example.com 104.18.27.120
2026-03-13T16:10:56.442608-0400 INFO A example.com 104.18.26.120
2026-03-13T16:10:56.443084-0400 INFO AAAA example.com 2606:4700::6812:1a78
2026-03-13T16:10:56.443671-0400 INFO AAAA example.com 2606:4700::6812:1b78
2026-03-13T16:10:59.653930-0400 INFO TXT example.com v=spf1 -all
2026-03-13T16:10:59.655616-0400 INFO TXT example.com _k2n1y4vw3qtb4skdx9e7dxt97qrmmq9
2026-03-13T16:10:59.726640-0400 INFO TXT _dmarc._domainkey.example.com v=DKIM1; p=
2026-03-13T16:10:59.727716-0400 INFO Enumerating SRV Records
2026-03-13T16:11:03.167934-0400 ERROR No SRV Records Found for example.com
2026-03-13T16:11:03.168882-0400 INFO Completed enumeration for domain: example.com
```

16. Step 3 - Attempt DNS Zone Transfer

-In this step a command was entered that caused the replications to be transferred between dns servers.

```
(kali㉿kali)-[~]  
$ dig axfr example.com @192.168.56.102  
; <<>> DiG 9.20.18-1-Debian <<>> axfr example.com @192.168.56.102  
;; global options: +cmd  
; Transfer failed.
```

17. Step 4 -Advanced DNS Enumeration

- In this step a command was entered that was used to discover hidden subdomains.

```
(kali㉿kali)-[~]  
$ dnsrecon -d example.com -D /usr/share/wordlists/dnsmap.txt -t brt  
2026-03-13T16:16:27.203534-0400 INFO Using the dictionary file: /usr/share/wordlists/dnsmap.txt (provided by user)  
2026-03-13T16:16:27.204199-0400 INFO Starting enumeration for domain: example.com  
2026-03-13T16:16:27.204802-0400 INFO brt: Performing host and subdomain brute force against example.com...
```